



DPUSCNXHedgehogKiller

Road network dependant control of other DPUs.

1. Parameters:

<i>Family</i>	Family of the hedgehogs of this instance of the DPU.
<i>Name[1...10]</i>	Hedgehog name for which the respective outputs are <i>Out1</i> to <i>Out10</i> are responsible.
<i>Default[1...10]</i>	Initial values for the outputs <i>Out1</i> to <i>Out10</i>
<i>OutMode[1...10]</i>	Switching behaviour of the output
<i>OutMin[1...10]</i>	Minimum value of the respective output
<i>OutMax[1...10]</i>	Maximum value of the respective output

2. Outputs:

<i>Out[1...10]</i>	Output values that will be set if the EGO-vehicle drives over a hedgehog.
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3. Functional principle:

The DPUSCNXHedgehogKiller can be used to control inputs of DPUs depending on the EGO-vehicles position in the road network of the database SCNX (s. DPUSCNX). Therefor at certain positions of the road network events – s.c. “Hedgehogs” – are placed. If the EGO-vehicle drives over such a Hedgehog the outputs of DPUSCNXHedgehogKiller are set accordingly. By connecting these outputs to in-puts of other DPUs these DPUs can be controlled depending on the drivers position in the road network. Detailed information on defining Hedgehogs can be found in the documentation „Reference Database SCNX.pdf“.

Hedgehog definition in the road network definition:

```
Hedgehogs =  
{  
    (<position along the road>, <family>  
    ,<tokenX>,<valueX>  
    ,<tokenY>,<valueY>  
    ,<tokenZ>,<valueZ>  
    ,...  
    )  
};
```



The DPUSCNXHedgehogKiller has 10 outputs *Out1* to *Out10*. All outputs are categorized into a s.c. family. The name of this family as well as the names for the single parameters can be chosen freely. When driving over a Hedgehog of the example above the output *Out<index>*, whose name in *Name<index>* has been defined as „tokenX“, becomes „valueX“. This value is kept until the driver drives over another Hedgehog with the name *Name<index>*. At start of the simulation *Out<index>* has the default value define in *Default<index>*.

If more than 10 outputs for a certain family are needed, multiple instances of the DPUSCNXHedgehogKiller can be created.

The switching behaviour of *Out<index>* is controlled by the parameter *Out-Mode<index>*. A tuple is assigned to this parameter:

```
OutMode<index> = (<mode>,<param>);
```

OutMode	Description	Param1
Set	The output value is set and kept.	-
Add	The new value is added to the current value of the output.	-
MonoFlop	The output value is kept for a time Param1 ms and set to 0 afterwards.	Zeit
String	The value of the Hedgehog is interpreted as string. In this case the parameters <i>OutMin<index></i> and <i>OutMax<index></i> have no effect in this case, since numerical comparisons are undefined for strings.	-

Example of a DPU configuration

```
...
DPUSCNXHedgehogKiller AccControl
{
    Computer = {"ASS"};
    Index = 10;
    Family = "Acc";
    Name1 = "TargetVelocity";
    Default1 = 0.0;
    Name2 = "TargetHeadway";
    Default2 = 1.5;
};
```

```
DPUAcc Acc
{
```



SILAB DRIVING SIMULATION

version Version 7.1 documentation

```
    Computer = {"ASS"};
    Index = 20;
    TargetVelocity = 0.0;
    TargetHeadway = 1.5;
};
Connections =
{
    AccControl.Out1 <-> Acc.TargetVelocity,
    AccControl.Out2 <-> Acc.TargetHeadway
};
```

In this example the inputs *TargetVelocity* and *Headway* of the DPUAcc are controlled by Hedgehogs.

The family „Acc“ is used. Only parameters defined within this family will be processed. The first output *Out1* is responsible for the parameter with name „TargetVelocity“. Therefore, *Out1* reacts on the parameter *TargetVelocity* in the Hedgehog definition. The second output *Out2* is responsible for *TargetHeadway*. The instance *AccControl* of *DPUSCNXHedgehogKiller* is connected to the Acc DPU afterwards. The inputs *TargetVelocity* and *TargetHeadway* are connected with the corresponding outputs of DPU *AccSteuerung*.

Example for the Hedgehog definition in the road network:

```
...
Hedgehogs =
{
    (... ,Acc,TargetVelocity,27.667,TargetHeadway,1.7) ,
    (... ,Acc,TargetVelocity,13.889) ,
    (... ,Acc,TargetHeadway,1.5)
}
```

If the driver drives over the first hedgehog the input *TargetVelocity* is set to 100 km/h (27.667 m/s) and the input *TargetHeadway* to 1.7s. The second line only sets Targetvelocity (*Out1*) to 50 km/h. The third line only sets TargetHeadway.